

# Math Calculations

$$v = \sqrt{19.6d}$$

$0.9 < d < 3$  meters. This converts to approx 3 to 10 feet.

$4.29 < v < 7.63$  meters per second.

Where  $v$  is the initial velocity required.

$D$  is the distance in meters that the robot will jump on earth.

# Requirements

**Primary Objective:** To design, build, and test a lightweight, durable, and reusable jumping robot. The robot must be capable of taking high-quality aerial images of the lunar surface, surviving the landing, and autonomously resetting itself for subsequent jumps.

Want to make a controlled linear actuation system

We are converting energy to linear motion to propel the robot upwards.

Linear elastic actuator

Power modulated translational system

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## Key Goals & Performance Metrics

- **Survivability:** The robot's body and all components must withstand the shock of repeated landings from heights of up to 10 feet on Earth (~60 feet on the Moon).
  - **Autonomy:** The robot must be able to **self-right** and **reset its jumping mechanism** without any external assistance after landing.
  - **Jump Performance:**
    - Achieve a consistent jump height between **3 to 10 feet**.
    - Minimize the **time required to reset** between jumps.
    - Maximize the **total number of jumps** before requiring maintenance or repair.
  - **Imaging & Orientation:**
    - Capture clear, stable images of the ground during ascent and descent using a high-shutter-speed camera.
    - Maintain a consistent **camera orientation** (pointed downwards) throughout the jump.
  - **Control & Data Transfer:** All systems (jump motor, release mechanism, camera) must be wirelessly controlled. The robot will store photos and transmit them via **Bluetooth** to a nearby device when a connection is established.
  - **Power System:** The robot will be powered by a battery recharged by **solar panels**. The panels must be positioned to avoid damage during landing and to effectively charge in the low-angle sunlight of the lunar south pole.
  - **Logistics & Safety:**
    - Integrate a **locator beacon** (e.g., a flashing light) to make the robot easy to find after a jump.
    - The jump release mechanism must be reliable, preventing premature activation.
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## Operating Environment (Lunar South Pole)

- **Terrain:** The robot will operate on **hilly, uneven, and rocky terrain**, which will affect landing stability and the direction of subsequent jumps.
  - **Lighting:** Constant **low-angle sunlight** will create long, high-contrast shadows, which is a key consideration for camera exposure settings and solar panel charging.
  - **Atmosphere:** The robot must be designed for a **vacuum**. Aerodynamic elements like fins or parachutes are ineffective and should not be used.
  - **Temperature:** Materials, especially for energy storage (springs), must be chosen to function reliably across the **extreme temperature ranges** of the Moon.
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## Design Constraints & Limitations

- **Launch Delay:** The robot must have a built-in **10-20 second delay** between being set down and initiating its first jump.
- **Stored Energy:** The amount of potential energy stored in the jumping mechanism is inherently limited by the 10-foot maximum jump height.
- **Communication:** Data transfer is limited to **short-range Bluetooth**, requiring a rover or astronaut to be in proximity to retrieve images.
- **Safety:** The design must have **no sharp external parts**.

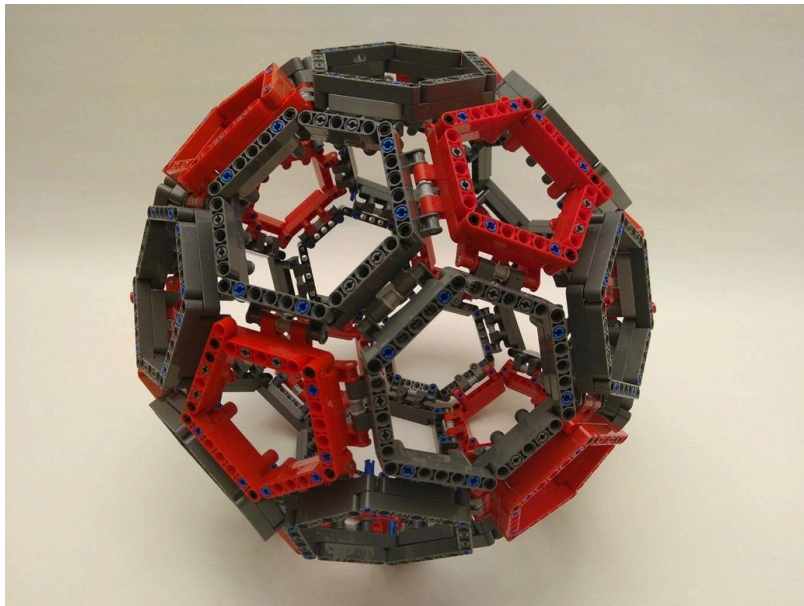
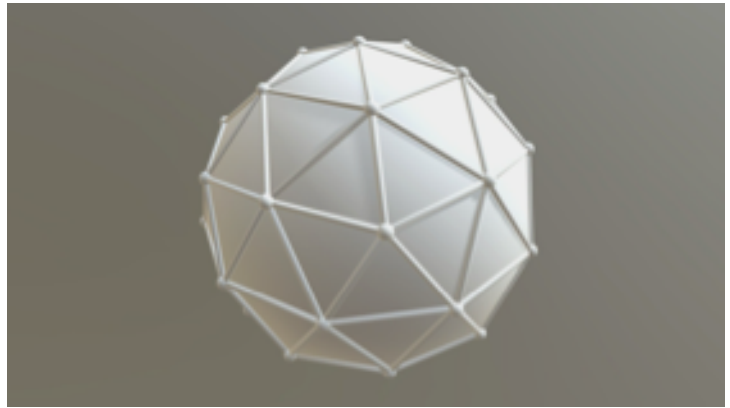
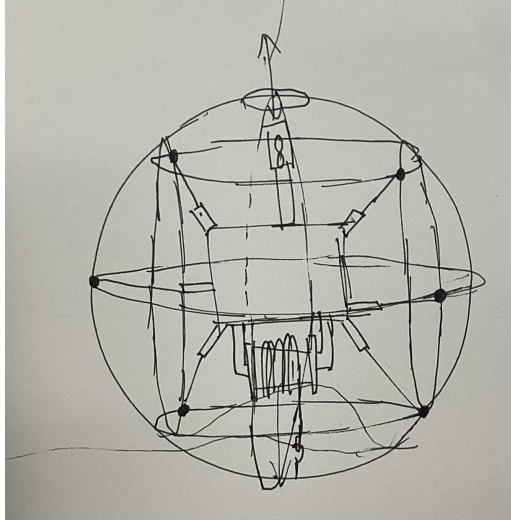
# Main Design

# 1st Concept Design Overview

Size: 10 inches diameter

The proposed jumping mechanism operates by using an internal **winch** to wind a cable, which pulls a section of the sphere's flexible frame inward. This action stores significant potential energy, much like a popper toy. The frame itself is a geodesic structure composed of **triangle panels** outlined with **spring steel** to allow for this elastic deformation. When the winch rapidly releases the cable, the inverted frame section violently snaps back to its original convex shape, launching the sphere into the air.

For landing, a network of internal, slightly elastic cables provides **cushioning**. These cables are attached to the outer shell and kept taut by a solenoid, creating a shock-absorbing suspension system. After landing, an autonomous **self-righting system**, using solenoids attached to a central control unit, reorients the sphere for subsequent operations. Internally, a **camera is suspended** to maintain a stable view, and the winch hook is permanently secured to the sphere's base with a robust, interlocking circular linkage (or we can make it very magnetic and have it attached to different points). Has a flashing mechanism attached for ease of locating.



## Material Components

Outershell:

Inner shell material ideas: aluminum,

Inner mechanisms: gyroscope,





# Shopping List

- Spring steel sheets-- 1075 steel
- 4 solenoids
- Cable for winch
- Winch and motor
- Gears for MA
- Inertial measurement unit
- Camera and camera solenoids
- Aluminum rods
- Frame materials

## Shopping list for NASA Design Team

- Spring steel sheet
  - 4 solenoids
  - Cable for winch
  - Winch & motor
  - Gears for MA
  - Inertial measurement unit
  - Camera + camera solenoids
  - Aluminum rods & frame materials
- } First prototype

Ebay  
HFI-B6 6 Axis ROS Robot  
Imu Module Arhs Attitude...

M5Stack-Store  
6-Axis IMU Unit(MPU6886)  
| m5stack-store

daischsensord  
High Precision 6-axis IMU  
IMIR

Adafruit  
Adafruit ISM330DHCX - 6  
DoF IMU - Accelerometer...

Most 6-axis IMUs (which combine a 3-axis accelerometer and a 3-axis gyroscope) have a wide price range depending on quality, precision, and intended application.

#### Typical Price Ranges

- **Low-cost DIY and robotics modules:**  
Common options like the MPU-6050 or MPU-6886 cost between \$3 and \$10 USD. These are ideal for small robotics or hobby projects such as your jumping robot. [m5stack #1](#)
- **Mid-range precision sensors:**  
Industrial-grade modules like the Adafruit ISM330DHCX or HFI-B6 range from \$20 to \$50 USD, offering better stability, calibration, and temperature compensation. [ebay #1](#)
- **High-end professional IMUs:**  
Sensors like the Daisch IMIR or Firepower IMU115 cost between \$250 and \$600 USD and are used in drones, autonomous vehicles, and aerospace applications. [daischsensord #1](#)

#### Recommended for You

For a jumping robot, a low-cost 6-axis IMU (MPU-6050 or MPU-6886) is ideal — it provides sufficient accuracy for detecting orientation and acceleration at a fraction of the cost (under \$10).

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#### Related

Ask a follow-up

# Problems

## Current Design Problems:

- Metal Fatigue/Cold Brittleness
- Winch mechanism needs to have a quick release
- Cables can get really messy and difficult to keep within the structure
- Solenoids are super heavy and limited travel
- High chance of ripping cables and solenoids cuz inertia
- Camera mechanism might be faulty
- Keeping it upright might be difficult
- 

Improvements: put camera on a 3 axis gimbal,

Things to address: cable management, aiming mechanism, durability, solar power

battery, bending material and how to retain the material strength, cable length and reset time,

# Research Categories

# Research Categories

- **Energy Storage and Jumping Mechanisms**
- **Landing Buffering & Stability Mechanisms**
- **Autonomous Self-Righting and Reset Systems**
- **Strong, Lightweight Materials Selection**
- **Camera and Imaging Systems**
- **Data Storage and Wireless Transfers**
- **Solar-recharged Power System**
- **Locator and Beacon Systems**
- **Timing and Control Sequences (including delays)**



# Research

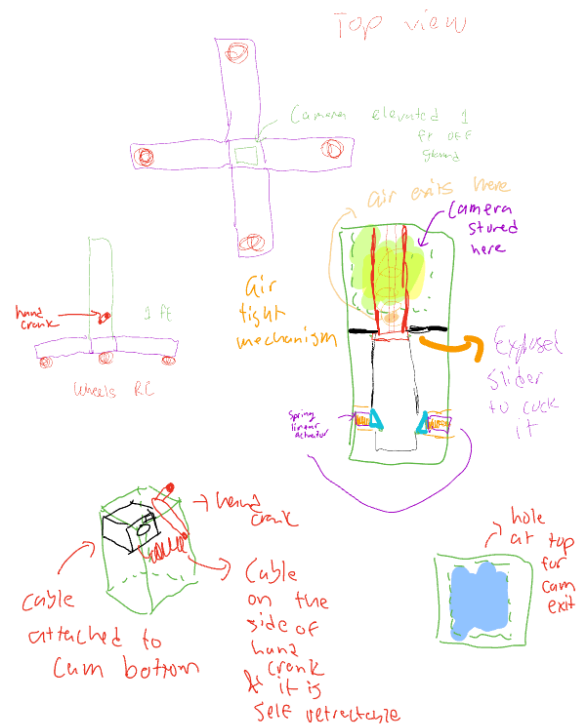
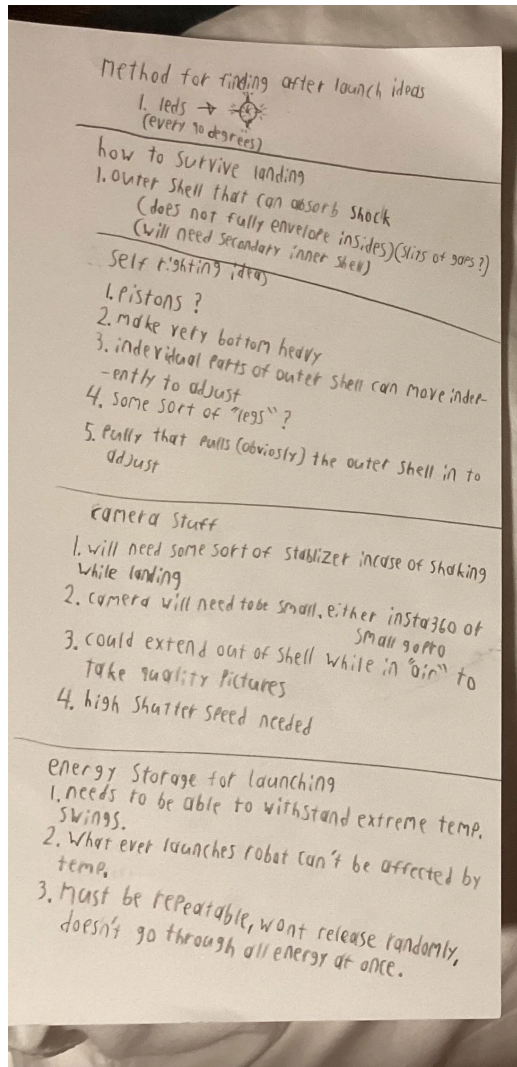
Solenoids- EMF is created from tightly wound coils of a conductive material which can induce linear motion through the activation of EMFs. Spring returns the piston back in place.

# Random Ideas and pics

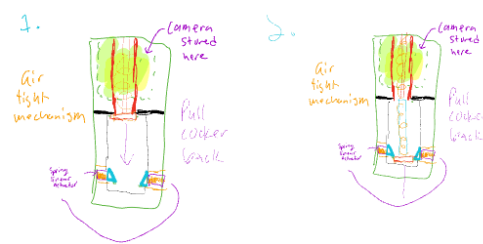
# Random images and ideas

- Symmetrical winch
- Pumpkin design
- Shock absorber system
- Inner Pendulum driven spherical robot
- Piston system
- No cable? no solenoid maybe?
- Spring system
- Gyroscope
- Rubber ball design
- There's nothing on the website saying to have the solar panels attached to the ball at all times, we could have solar panels sit separately then plug in when needed.
- Separate solar panels would also help reduce weight

# More rand images and ideas



## Mechanism



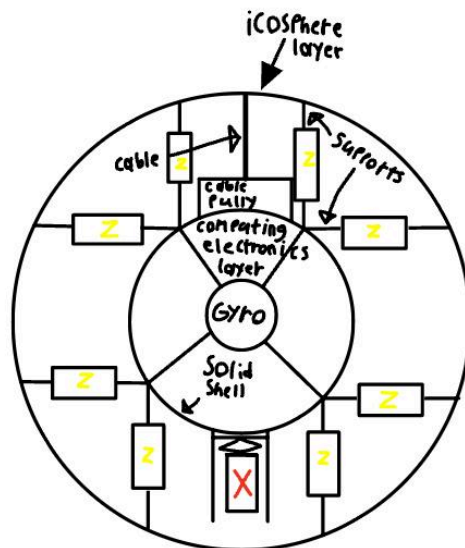
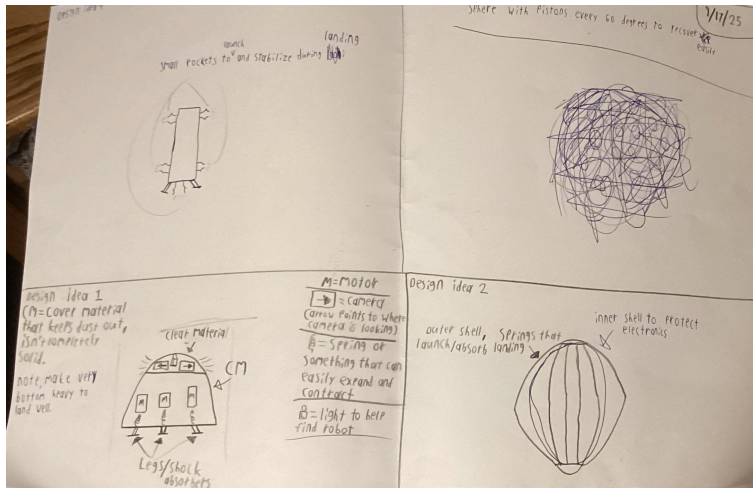
## 2. Release spring, actuator



## 3. Cam pushed upwards



## 4. Reel it back in w auto reel or manually



X = camera, will need stabilizer.  
will poke through hole in icosphere.  
will also need a piston or something like  
that to move it

Z = shock absorbers to protect internals  
from shaking

the supports are just to keep the internals  
in place.

Maybe make the Solid Shell  
out of aluminum

Have shock absorbers that look something like this



**Use blinking LEDs like this to locate**

